Ch 2. Presenting Data

- Frequency tables and Charts
- Bar Charts
- Stem and Leaf Displays
- Histograms
- Box Plots
- others

- Presenting qualitative/discrete data:

Previous Ownership	Frequency	Relative Frequency
None	85	0.17
Windows	60	0.12
Macintosh	355	0.71
Total	500	1.00

Frequency table

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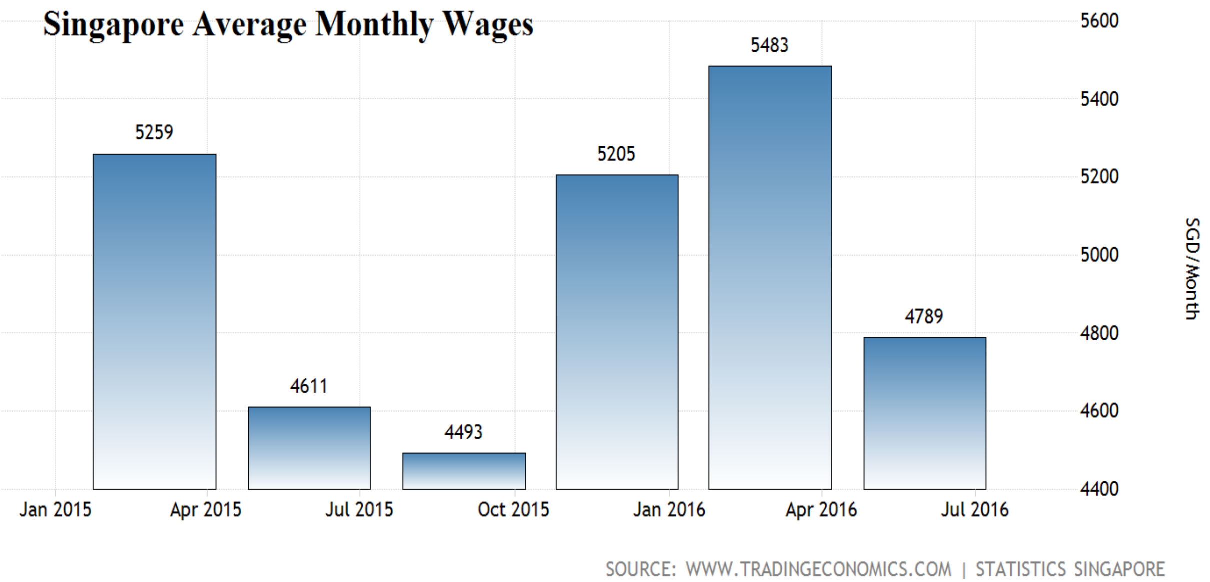
Frequency table



Bar chart

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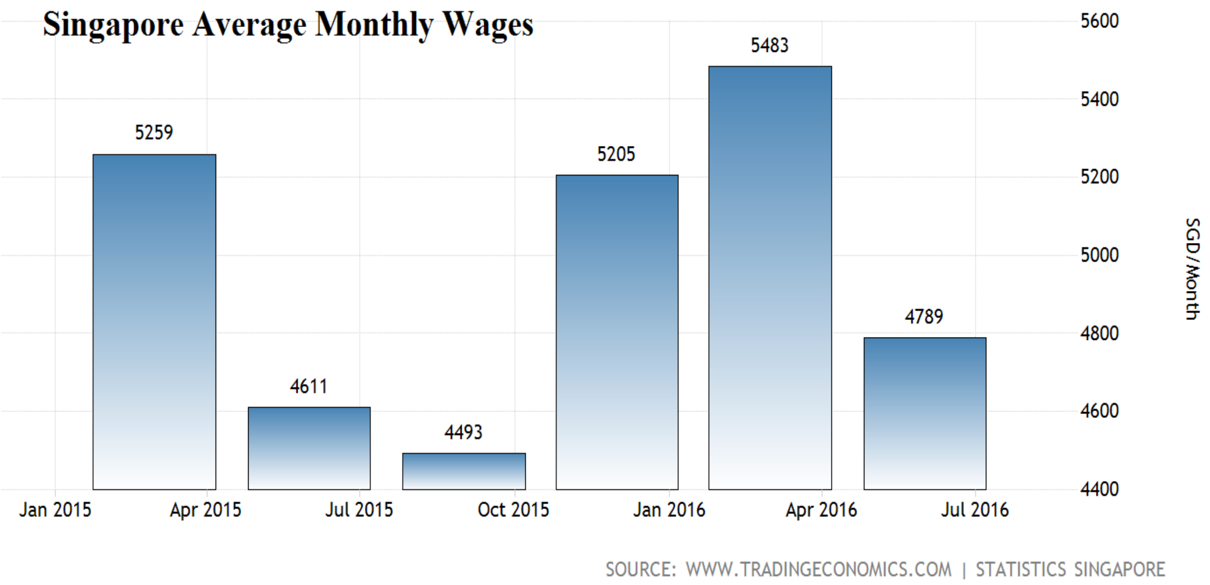
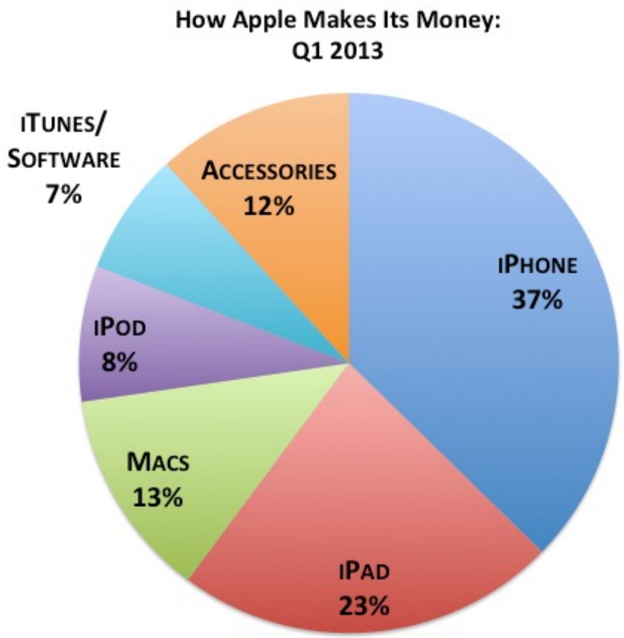
Frequency table

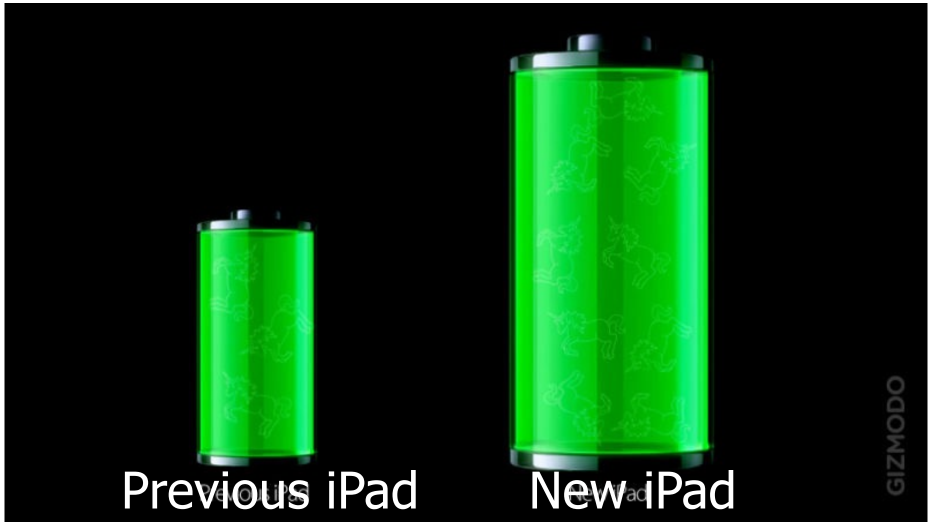




Bar chart

Pie chart

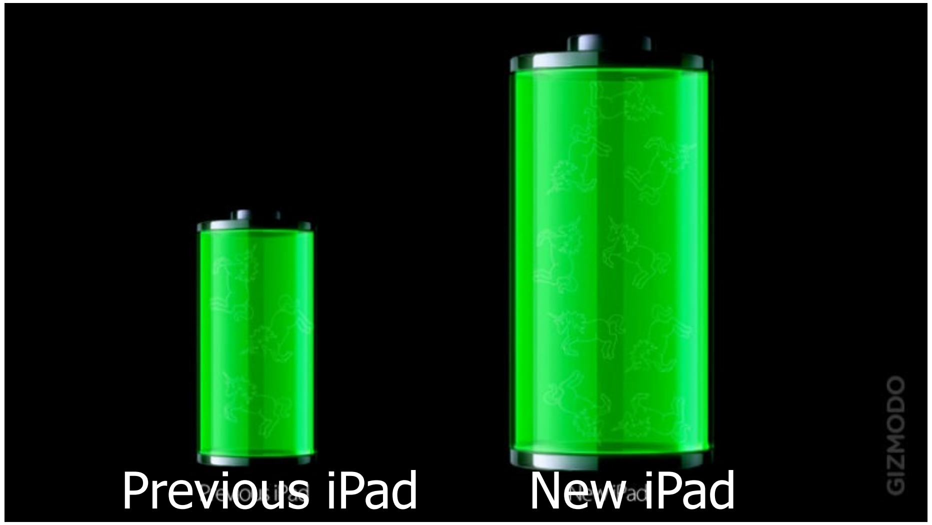
Examples of misleading presentations:



The battery on the right has 70% more capacity than the one on the left.

Misleading display?

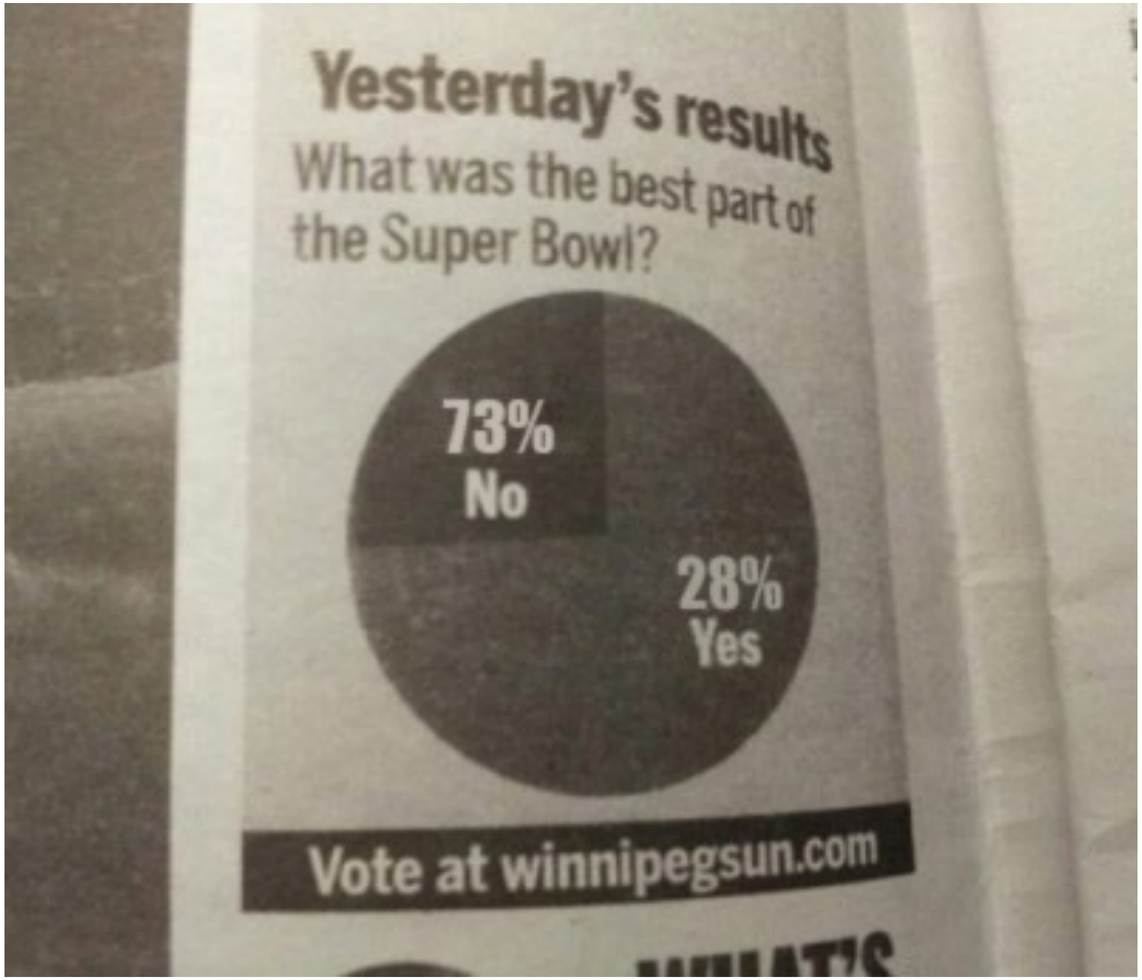
Examples of misleading presentations:



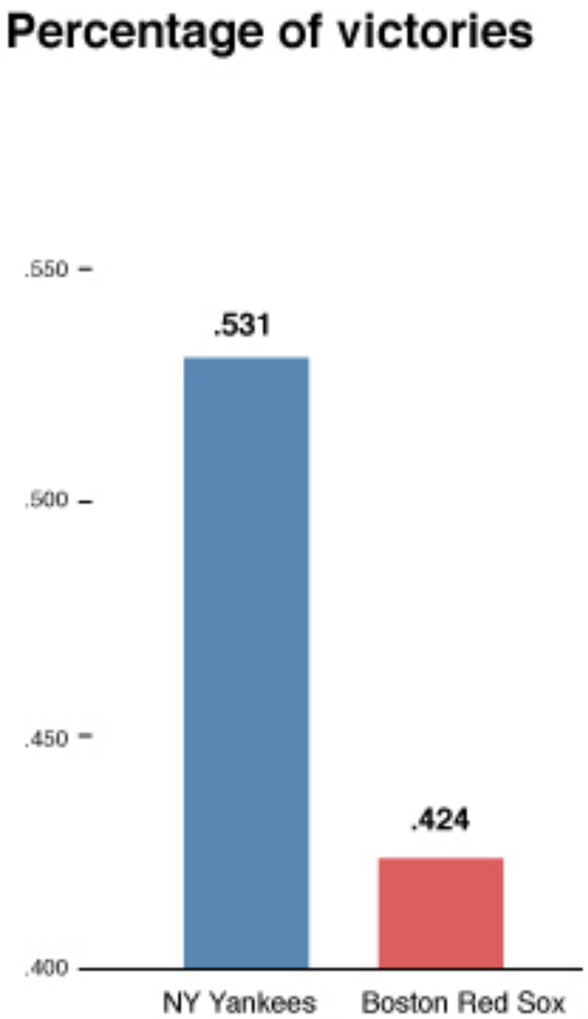
Yes!

The area of the battery on the right is about 3x the one on the left.

Examples of misleading presentations:



Examples of misleading presentations:



Is this misleading?

- Stem-and-leaf Presentation

- Useful for small datasets

Eg: No. of touchdown passes by 31 football teams.

37, 33, 33, 32, 29, 28, 28, 23, 22, 22, 22, 21, 21, 21, 20, 20, 19, 19, 18, 18, 18, 18, 16, 15, 14, 14, 14, 12, 12, 9, 6

- Stem-and-leaf Presentation

- Useful for small datasets

Eg: No. of touchdown passes by 31 football teams.

37, 33, 33, 32, 29, 28, 28, 23, 22, 22, 22, 21, 21, 21, 20, 20, 19, 19, 18, 18, 18, 18, 16, 15, 14, 14, 14, 12, 12, 9, 6

Stem

3
2
1
0

- Stem-and-leaf Presentation

- Useful for small datasets

Eg: No. of touchdown passes by 31 football teams.

37, 33, 33, 32, 29, 28, 28, 23, 22, 22, 22, 21, 21, 21, 20, 20, 19, 19, 18, 18, 18, 18, 16, 15, 14, 14, 14, 12, 12, 9, 6

Stem

3 | 2337
2 | 001112223889
1 | 2244456888899
0 | 69

Leaf

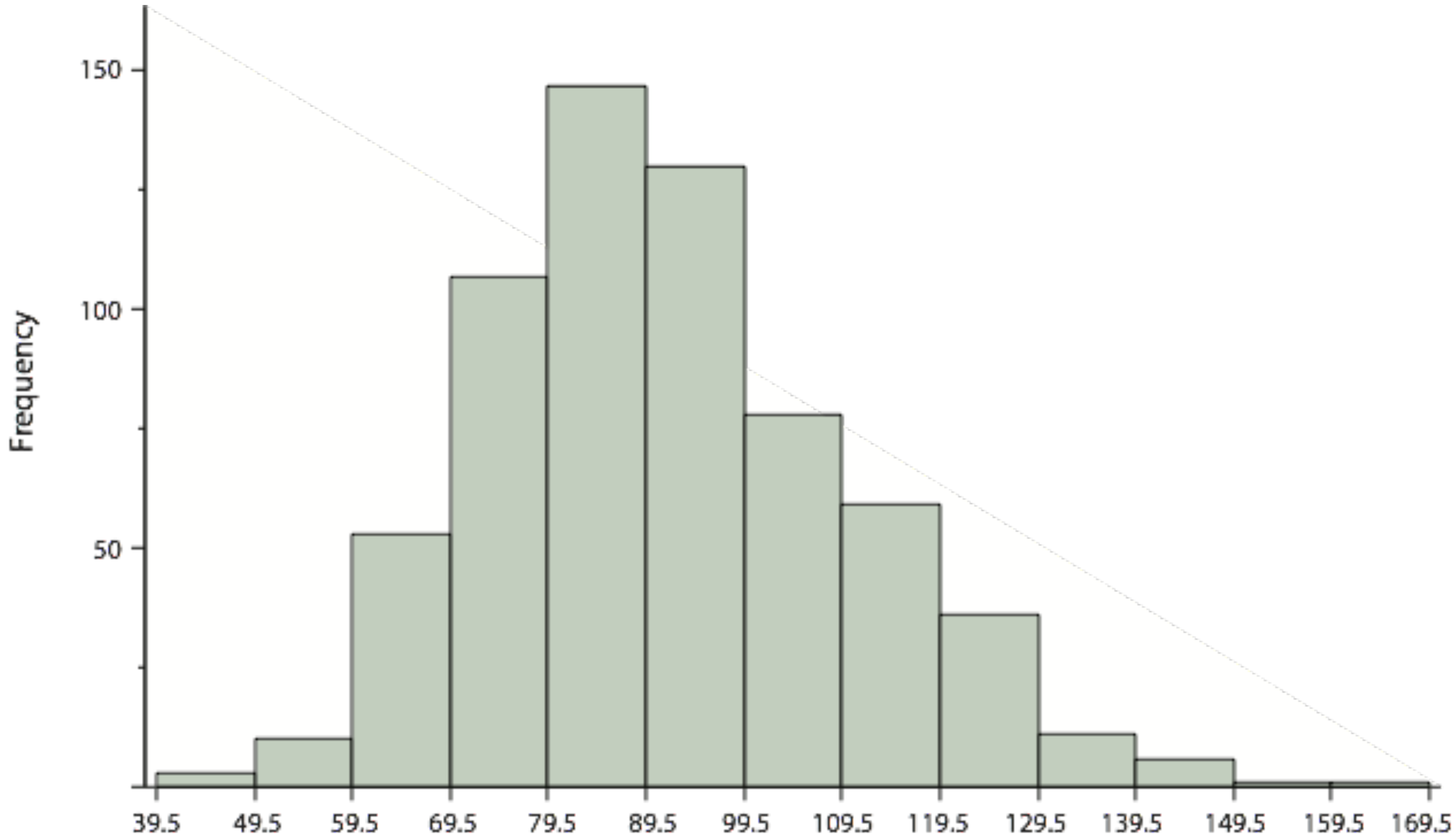
- Presenting continuous data:

  Range is divided into class intervals/bins

Interval's Lower Limit	Interval's Upper Limit	Class Frequency
39.5	49.5	3
49.5	59.5	10
59.5	69.5	53
69.5	79.5	107
79.5	89.5	147
89.5	99.5	130
99.5	109.5	78
109.5	119.5	59
119.5	129.5	36
129.5	139.5	11
139.5	149.5	6
149.5	159.5	1
159.5	169.5	1

- Presenting continuous data:

  Range is divided into class intervals/bins



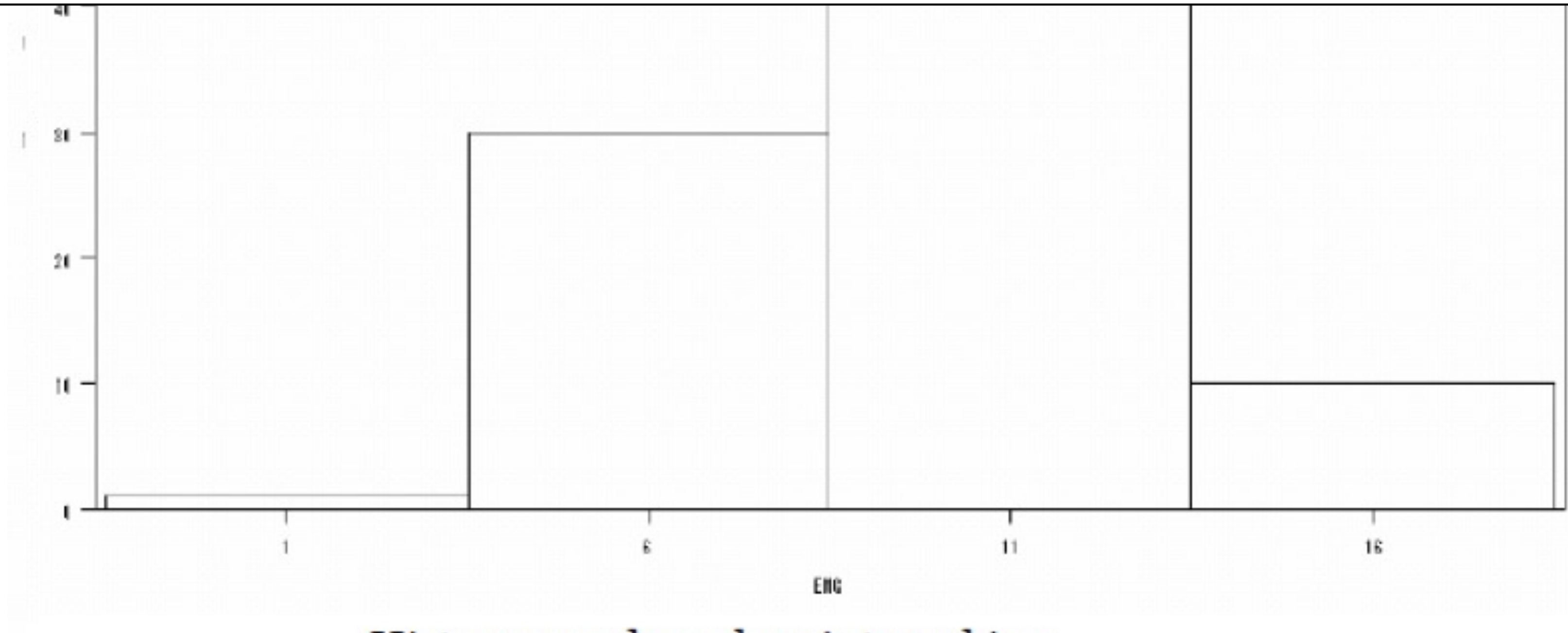
Histogram of the test scores

Presenting continuous data - Histogram:

Interval size affects the visual presentation

A rule of thumb:

$$\# \text{ of class intervals} \approx \sqrt{\# \; of \; observations}$$



Histogram when class interval is 5

Question: You have to decide between displaying data with a histogram or a stem-and-leaf display. What factor would affect your choice?

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For a given set of data, the choice is to select between a histogram or a stem-and-leaf display.

With more data, a histogram can be very useful since it shows the overall shape of the distribution.

Stem-and-leaf display is better for smaller sets of data.

Suppose you are constructing a histogram for describing the distribution of salaries for individuals who are 40 years or older, but are not yet retired.

a) What is on the X-axis?
b) What is on the Y-axis?
c) What would the shape of the salary distribution be like?

a) Salary would be on the X-axis because this is the variable whose distribution is of interest.

b) The Y-axis would be the frequency of individuals.

c) The distribution is expected to be positively skewed with a few wealthy CEOs earning well above the average salary.

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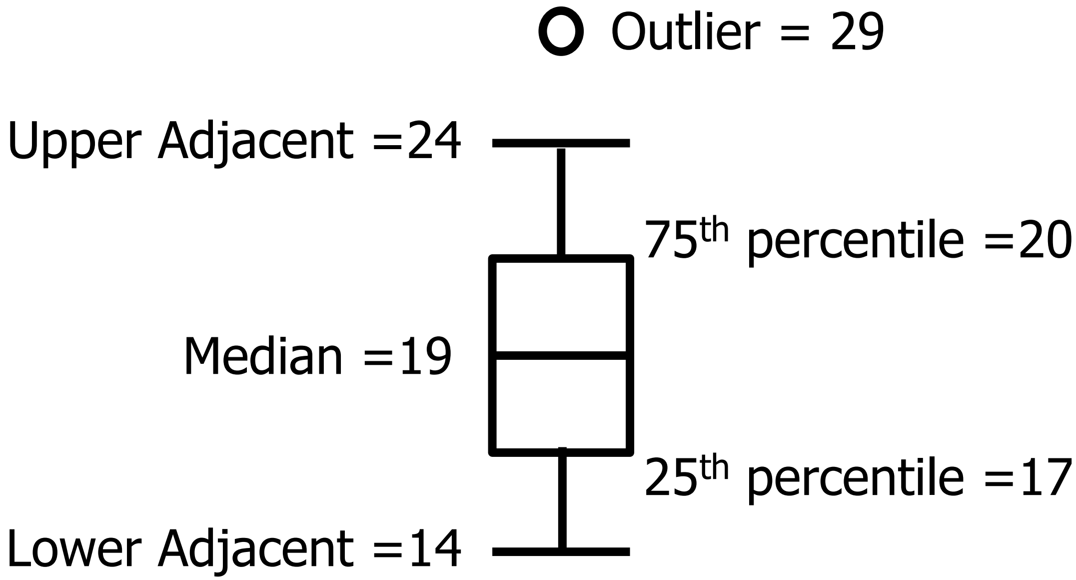
- Box Plot

- Identifying extreme outliers
- Summarising and comparing distributions



Draw a box plot for the following data set:

Data Set: 31 values

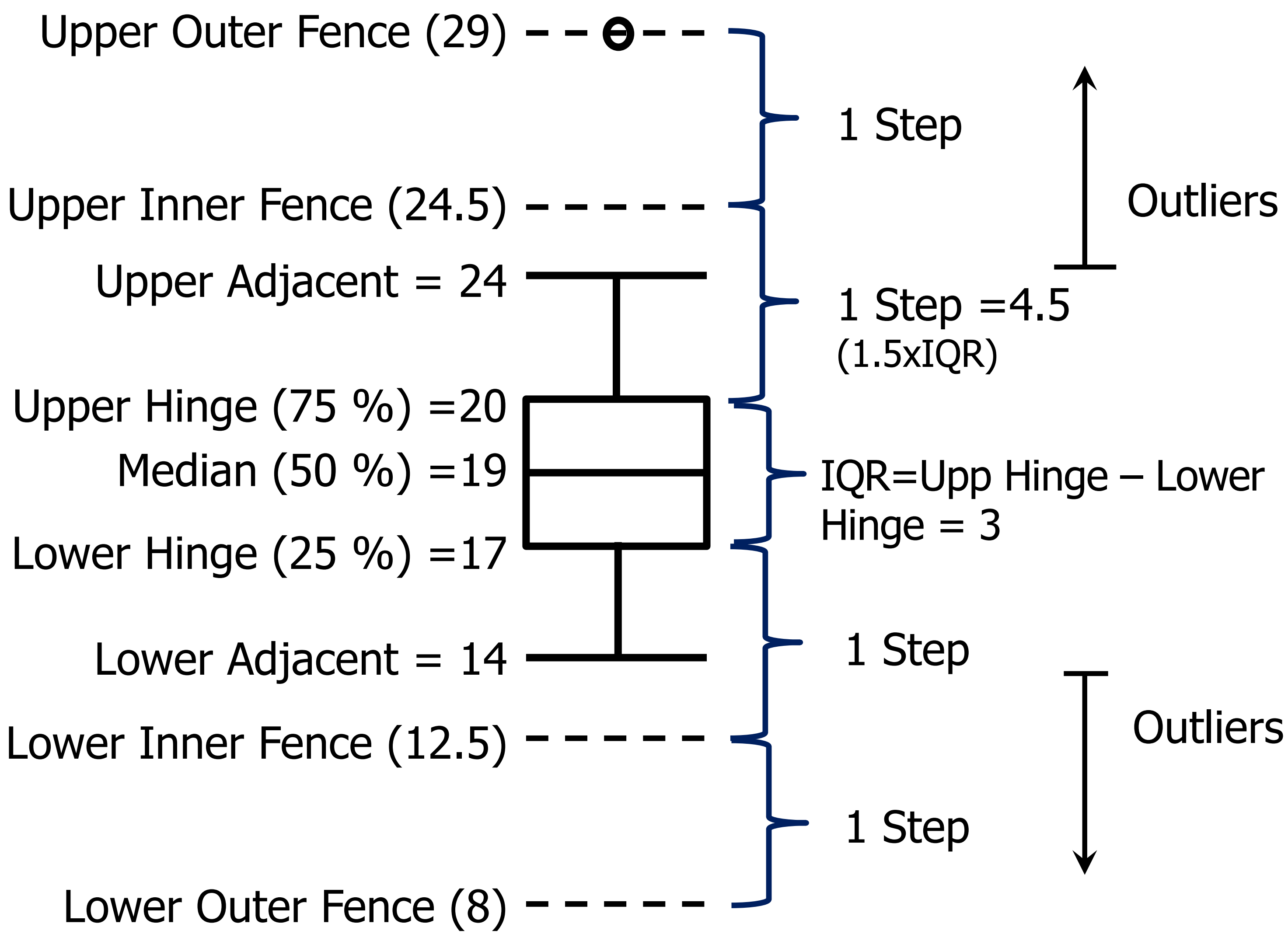
14	15	16	16	17
17	17	17	17	18
18	18	18	18	18
19	19	19	20	20
20	20	20	20	21
21	22	23	24	24
29				

Draw a box plot for the following data set:

Data Set: 31 values

14	15	16	16	17
17	17	17	17	18
18	18	18	18	18
19	19	19	20	20
20	20	20	20	21
21	22	23	24	24
29				

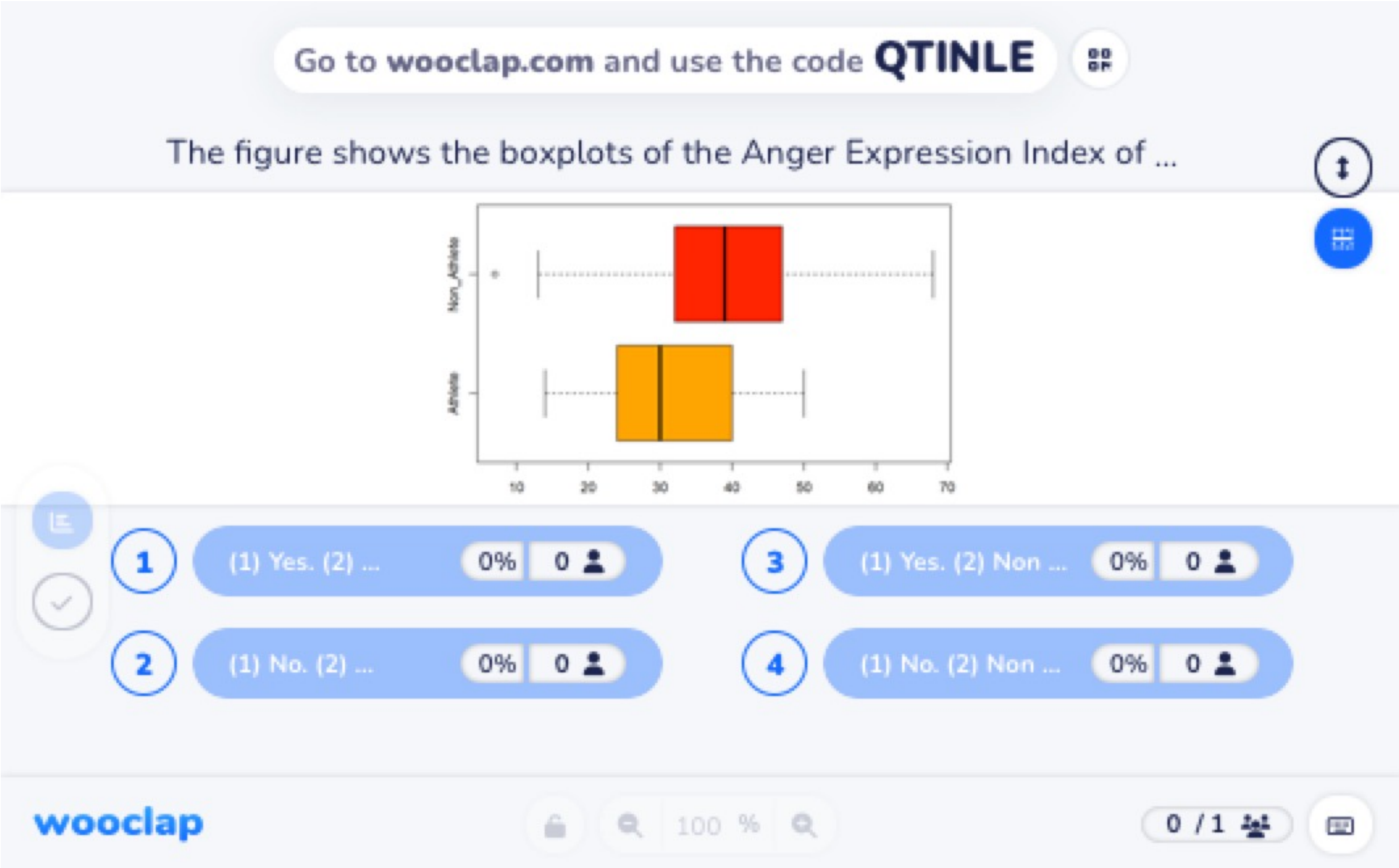
1. Cal 75^{th} and 25^{th} percentile
2. Determine the upper and lower inner fence thresholds
   – cal the step size = 1.5 x IQR
3. Determine the upper and lower outer fence thresholds
4. Obtain the upper and lower adjacent values
5. Label all outliers (values beyond the adjacent values)
6. Draw lines joining adjacent and hinge
7. Indicate the 50^{th} percentile (median)



Data Set: 31 values

14	15	16	16	17
17	17	17	17	18
18	18	18	18	18
19	19	19	20	20
20	20	20	20	21
21	22	23	24	24
29				

Figure shows the boxplots of the Anger Expression Index by sports participation. Does it look like there are any outliers? Which group reported expressing more median anger?

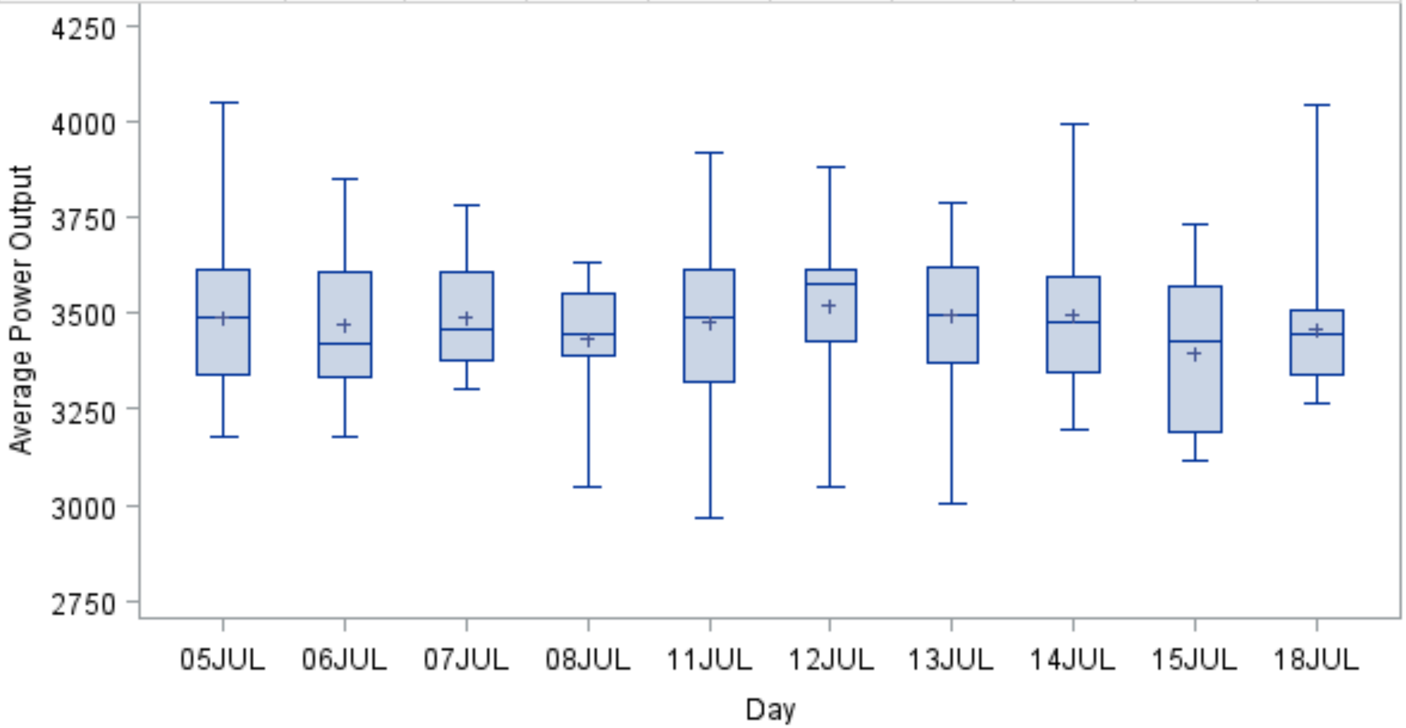


Power Output of an Electric Generator

Box Plot for Power Output

Overall Statistics			
Min	2968	Mean	3471.905
Max	4050	Pooled Std Dev	200.8349

Extremes by Day										
Min	3180	3179	3304	3045	2968	3047	3002	3196	3115	3263
Max	4050	3849	3781	3629	3916	3881	3787	3994	3731	4040



Daily share prices

Ch 3. Summarizing Distributions

- Central Tendency: mean, median & mode
- Other Measures of Central Tendency
- Comparing Central Tendency
- Measures of Variability: Range, IQR, Variance

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Central Tendency

Summarizes a distribution by its central location

Different ways to define central tendency:

- Mean $\mu = \frac{\sum X}{N}$
- Median = 50th Percentile
- Mode = the value with the highest frequency
- Trimean = $\frac{25^{\text{th}}\,\text{Percentile} + 2*\text{Median} + 75^{\text{th}}\,\text{Percentile}}{4}$
- Geometric mean $= (\prod X)^{\frac{1}{N}}$, where $\prod$ means to multiply

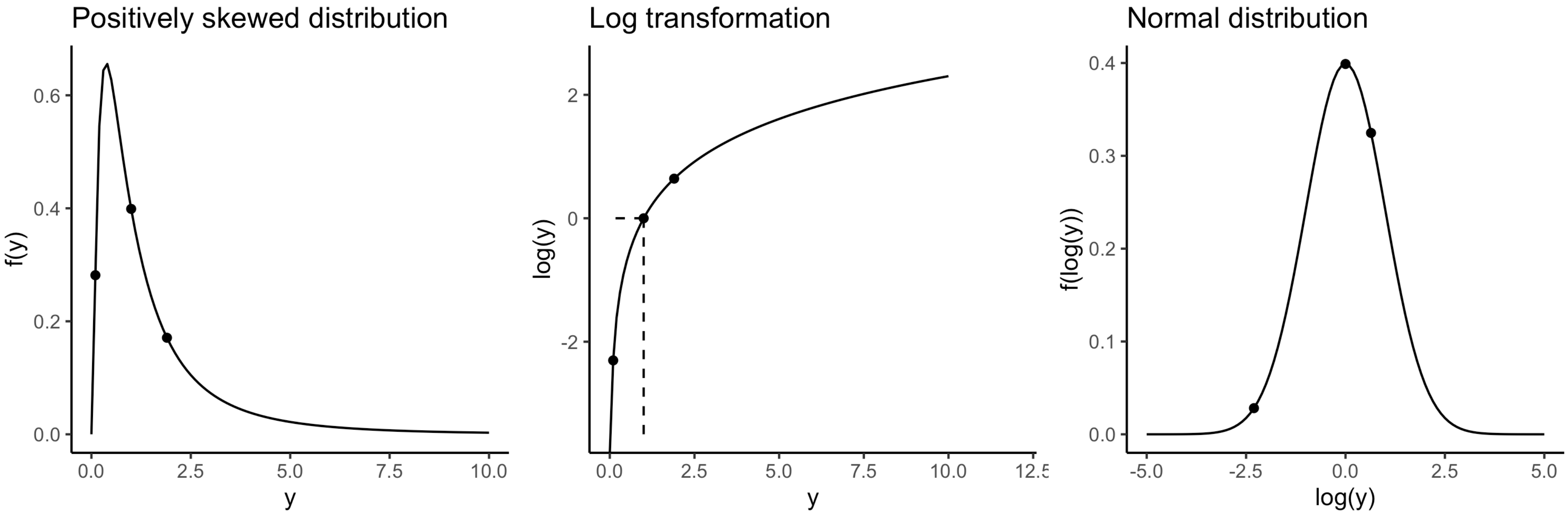
  Eg: $\prod_{i=1}^{5} X_i = X_1 * X_2 * X_3 * X_4 * X_5$
- Trimmed mean = mean for data with some higher and lower values removed

- Central Tendency

Eg: Given the following data set, compute the mean, the median, the mode, the trimean, the geometric mean and the mean trimmed 18.2%.

1	3	4	4	4	5	5	7	8	9	31

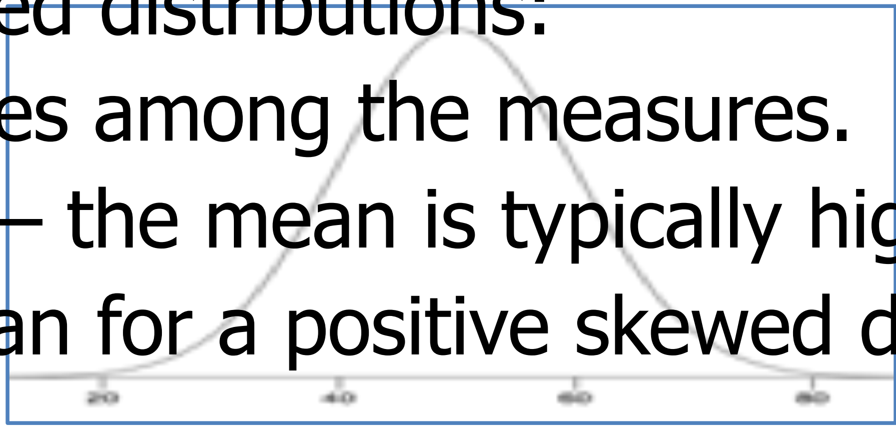
$$\text{Geometric mean} = \left(\prod X\right)^{\frac{1}{N}} = e^{\frac{1}{N}\sum \ln X}$$



- Central Tendency – comparing various measures

For symmetric distributions:
Mean = Median= Trimean = Trimmed mean = Mode (except bimodal distr)

For skewed distributions:
Differences among the measures.
Example – the mean is typically higher than the median for a positive skewed distribution



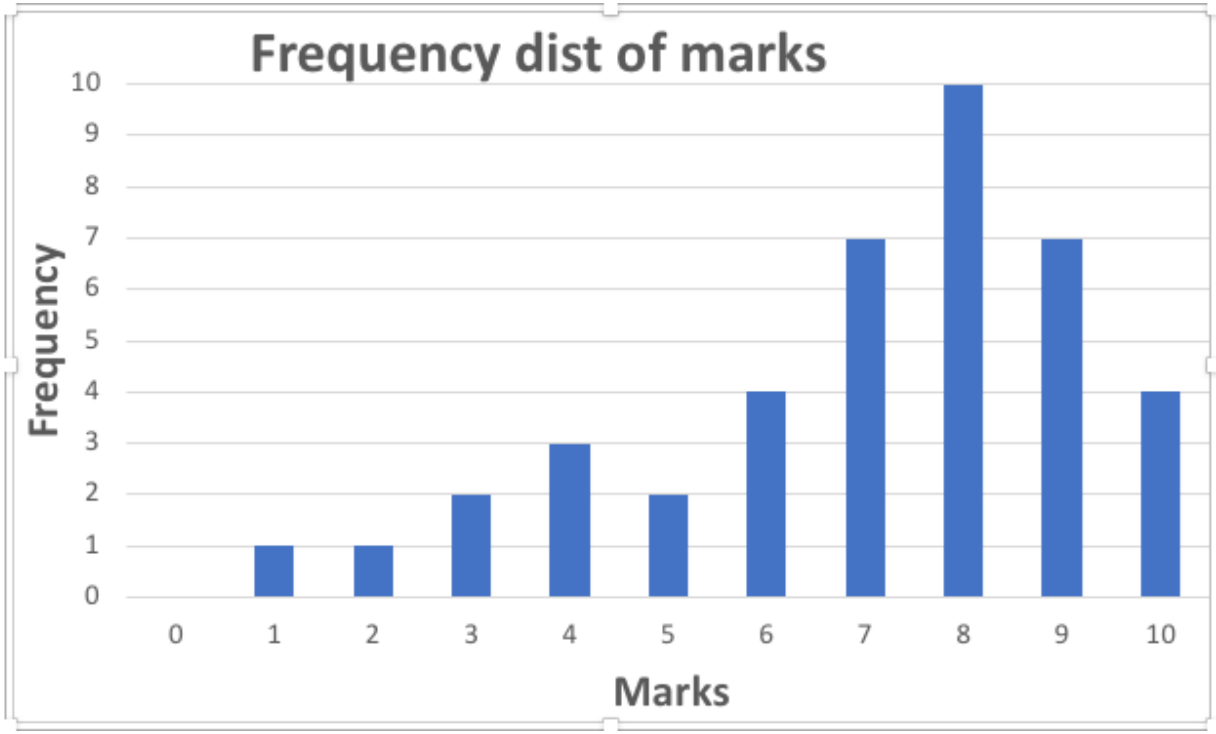
Symmetric

Central Tendency – comparing various measures Example

Negative skewed

Calculate the following:

1. mode
2. median
3. trimean
4. mean



- Measures of Variability

An indication of how spread out is the distribution

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An indication of how spread out is the distribution

Frequently used measures of variability:

- Range = Highest value – Lowest value
- Interquartile Range IQR = 75^{th} – 25^{th} Percentile
- Variance
- Standard deviation = $\sqrt{\text{Variance}}$

Measures of Variability

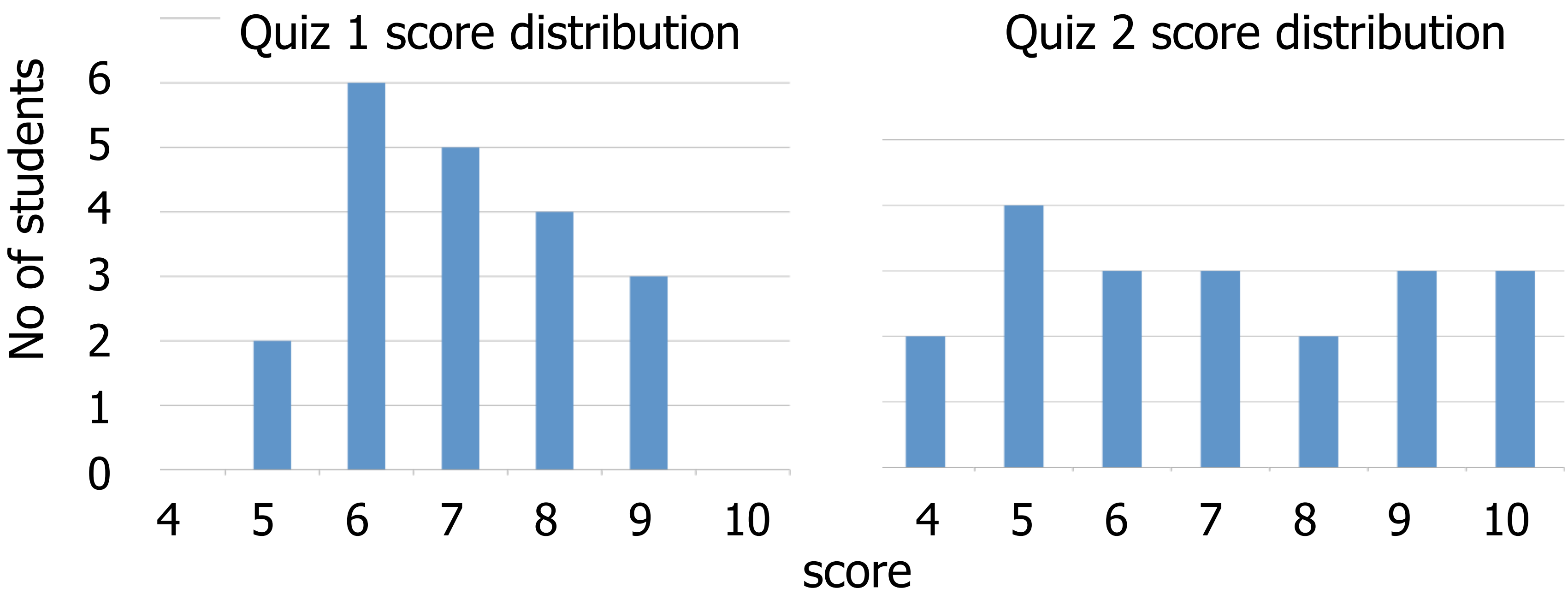
An indication of how spread out is the distribution

Frequently used measures of variability:

- Range = Highest value – Lowest value
- Interquartile Range IQR = $75^{th} - 25^{th}$ Percentile
- Variance $\sigma^2 = \dfrac{\sum(X-\mu)^2}{N}$
- Standard deviation = $\sqrt{\mathrm{Variance}}$

- Variability:

Eg: Given the following 2 data sets, compute the mean, the range, the IQR and the variance.



True or False questions:

Go to **wooclap.com** and use the code **QTINLE**

Enter T for True and F for false.

1. A bimodal distribution has two modes and two medians - 1
2. The best way to describe a skewed distribution is to report the mean - 2
3. When plotted on the same graph, a distribution with a mean of 50 and a standard deviation of 10 will look more spread out than will a distribution with a mean of 60 and a standard deviation of 5 - 3

wooclap

100 %

0 / 1

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Compute mean and variance

- From the **population** of size N:

Population Mean $\quad \mu = E[X] = \frac{\sum X}{N}$

Population Variance $\quad \sigma^2 = E[(X-\mu)^2] = \frac{\sum (X-\mu)^2}{N}$

Compute mean and variance

- From the **population** of size N:

Population Mean $$\mu = E[X] = \frac{\sum X}{N}$$

Population Variance $$\sigma^2 = E[(X-\mu)^2] = \frac{\sum (X-\mu)^2}{N} \quad \text{or} \quad \frac{\sum X^2 - \frac{(\sum X)^2}{N}}{N}$$

Compute mean and variance

- From the **population** of size N:

Population Mean $\quad \mu = E[X] = \frac{\sum X}{N}$

$$\boxed{\sigma^2 = E[X^2] - \mu^2}$$

Population Variance $\quad \sigma^2 = E[(X-\mu)^2] = \frac{\sum (X-\mu)^2}{N}$ or $\frac{\sum X^2 - \frac{(\sum X)^2}{N}}{N}$

Compute mean and variance

- From the **population** of size *N* :

Population Mean $\mu = E[X] = \frac{\sum X}{N}$

$\sigma^2 = E[X^2] - \mu^2$

Population Variance $\sigma^2 = E[(X-\mu)^2] = \frac{\sum(X-\mu)^2}{N}$ or $\frac{\sum X^2 - \frac{(\sum X)^2}{N}}{N}$

- From a **sample** of size *n* :

Sample Mean $\bar{x} = \frac{\sum X}{n}$

Sample Variance $s^2 = \frac{\sum(X-\bar{x})^2}{n-1}$ or $\frac{\sum X^2 - \frac{(\sum X)^2}{n}}{n-1}$

Suppose the denominator of s^2 is n, instead of $(n-1)$:

$$s^2 = \frac{\sum(X-\bar{x})^2}{n} = \frac{1}{n}\left(\sum X^2 - \frac{(\sum X)^2}{n}\right)$$

For unbiased estimate, we expect the mean of s^2 to be equal to σ^2 :

$$E[s^2] = E\left[\frac{1}{n}\left(\sum X^2 - \frac{(\sum X)^2}{n}\right)\right]$$

$$= \frac{1}{n}\left(\sum E[X^2] - \frac{E[(\sum X)^2]}{n}\right)$$

$\sigma^2 + \mu^2$ from the previous slide

$$= \frac{1}{n}\left(n\,\sigma^2 + n\,\mu^2 - \frac{E[(\sum X)^2]}{n}\right)$$

$$E[s^2] = \frac{1}{n}\left(n\,\sigma^2 + n\,\mu^2 - \frac{E[(\sum X)^2]}{n}\right)$$

Let $Y = \sum X$

$E[Y^2] = \sigma_Y^2 + \mu_Y^2$

$$= \frac{1}{n}\left(n\,\sigma^2 + n\,\mu^2 - \frac{Var[\sum X] + (E[\sum X])^2}{n}\right)$$

$$= \frac{1}{n}\left(n\,\sigma^2 + n\,\mu^2 - \frac{\sum Var[X] + (\sum E[X])^2}{n}\right)$$

$$= \frac{1}{n}\left(n\,\sigma^2 + n\,\mu^2 - \frac{n\,\sigma^2 + (n\,\mu)^2}{n}\right)$$

$$= \frac{1}{n}(n\,\sigma^2 + n\,\mu^2 - \sigma^2 - n\,\mu^2) \quad = \frac{1}{n}(n - 1)\,\sigma^2$$

If the denominator is $(n - 1)$, then the mean of s^2 is equal to σ^2, i.e. $E[s^2] = \sigma^2$

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- Linear transformation of variable

  - Transform data from one measurement scale to another

Eg. Consider a taxi trip from point A to B. The taxi service initial charge is \$3 and the trip costs an additional \$0.50 per km. Let y be the cost of the taxi ride and x be the distance travelled. We have:

$$y = 0.5x + 3$$

$$\boxed{y = ax + b}$$

The mean of y = 0.5 (mean of x) + 3

$$\boxed{E[y] = aE[x] + b}$$

- Linear transformation of variable

  - Transform data from one measurement scale to another

Eg. Consider a taxi trip from point A to B. The taxi service initial charge is \$3 and the trip costs an additional \$0.50 per km. Let *y* be the cost of the taxi ride and *x* be the distance travelled. We have:

$$y = 0.5x + 3$$

$\boxed{y = ax + b}$

The mean of y = 0.5 (mean of x) + 3

$\boxed{E[y] = aE[x] + b}$

The variance of y = 0.5^2 (variance of x)

$\boxed{Var[y] = a^2\, Var[x]}$

- Variance Sum Law I

  - Linear combination of 2 independent variables

Eg. Consider a couple A and B. A works *x* hrs/day and earns $5/hr while B works *y* hrs/day and earns $10/hr. Total daily income *t* is:

$$t = 5x + 10y$$

$$E[ax + by] = aE[x] + bE[y]$$

Mean = 5 (mean of *x*) + 10 (mean of *y*)

$$Var[ax + by] = a^2\,Var[x] + b^2 Var[y]$$

Variance = $5^2\ \sigma_x^2 + 10^2\ \sigma_y^2$

Variance Sum Law I

- Linear combination of 2 independent variables

Eg. Consider a couple A and B. A works *x* hrs/day and earns \$5/hr while B works *y* hrs/day and earns \$10/hr. Total daily income *t* is:

$$t = 5x + 10y$$

Daily income difference *d* :

$$d = 5x - 10y$$

$E[ax + by] = aE[x] + bE[y]$

Mean = 5 (mean of *x*) – 10 (mean of *y*)

$Var[ax + by] = a^2\,Var[x] + b^2 Var[y]$

Variance = $5^2\,\sigma_x^2 + 10^2\,\sigma_y^2$